

Air University



Digital Logic Design (Lab)

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Experiment No. 07

Title: Decoders and Encoders

Objectives:

- To understand the fundamental workings of encoders and decoders in digital electronics.
- To design and build decoder and encoder circuits using logic gates.
- To verify their effectiveness through truth tables and Proteus simulations.

Equipment and Components:

- Proteus Software (for simulation)
- Logic Trainer Kit (for hands-on practice)
- Logic Gate ICs (7408 - AND, 7404 - NOT)
- Breadboard
- Power Supply (5V)
- Connecting Wires
- Multimeter (optional)

Introduction:

Encoders and decoders play a crucial role in digital electronics as essential combinational circuits. A **decoder** takes an N-bit binary input and activates one out of 2^N possible output lines, depending on the input combination. These devices are commonly used in applications such as memory addressing, data routing, and code conversion. In contrast, an **encoder** performs the opposite function, converting 2^N input lines into an N-bit binary code. Encoders are often used in communication systems to efficiently package data for transmission. Both types of circuits help to simplify complex digital systems and facilitate effective signal processing.

In this experiment, the focus will be on designing and implementing decoder and encoder circuits using logic gates. We will confirm their functionality through truth tables and simulate their operations using Proteus.

Procedure:

Decoder Design and Simulation

1. Design:

- a. Create a 3-to-8 decoder circuit utilizing three inputs (A, B, C).
- b. Connect these inputs to AND gates along with NOT gates to generate all possible output combinations.

2. Implementation:

- Set up the required ICs (AND and NOT gates) on the Logic Trainer Kit or in the Proteus workspace.
- Connect inputs A, B, and C to the circuit, making sure to wire all eight output lines correctly.

3. Verification:

- Use Proteus to simulate the circuit, ensuring that only one output is HIGH for each input combination as outlined in the truth table.

Logic Expressions and Truth Tables

Decoder Logic Expressions:

The output lines are activated based on the combinations of inputs A, B, and C via AND gates; The complete logic expressions for a 3-to-8 decoder are derived based on the truth table, where three inputs (A, B, C) determine eight unique outputs (D0 to D7). Each output is activated for a specific combination of inputs. The logic expressions for the outputs are as follows:

- $D0 = \bar{A} \cdot \bar{B} \cdot \bar{C}$
- $D1 = \bar{A} \cdot \bar{B} \cdot C$
- $D2 = \bar{A} \cdot B \cdot \bar{C}$
- $D3 = \bar{A} \cdot B \cdot C$
- $D4 = A \cdot \bar{B} \cdot \bar{C}$
- $D5 = A \cdot \bar{B} \cdot C$
- $D6 = A \cdot B \cdot \bar{C}$
- $D7 = A \cdot B \cdot C$

Explanation:

Each output (D_x) is activated by a specific combination of inputs (A, B, C):

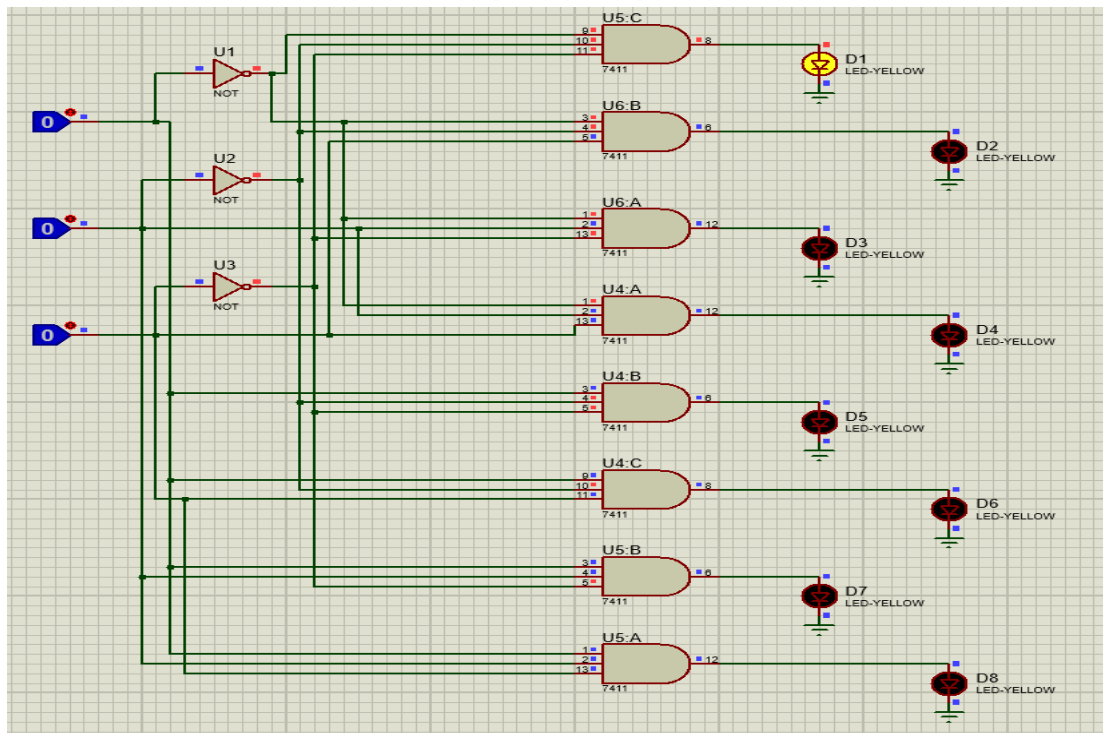
- The NOT gates are used to generate the complemented inputs ($\bar{A}$, $\bar{B}$, $\bar{C}$).
- AND gates combine the inputs and their complements to produce the desired outputs.

Truth Table:

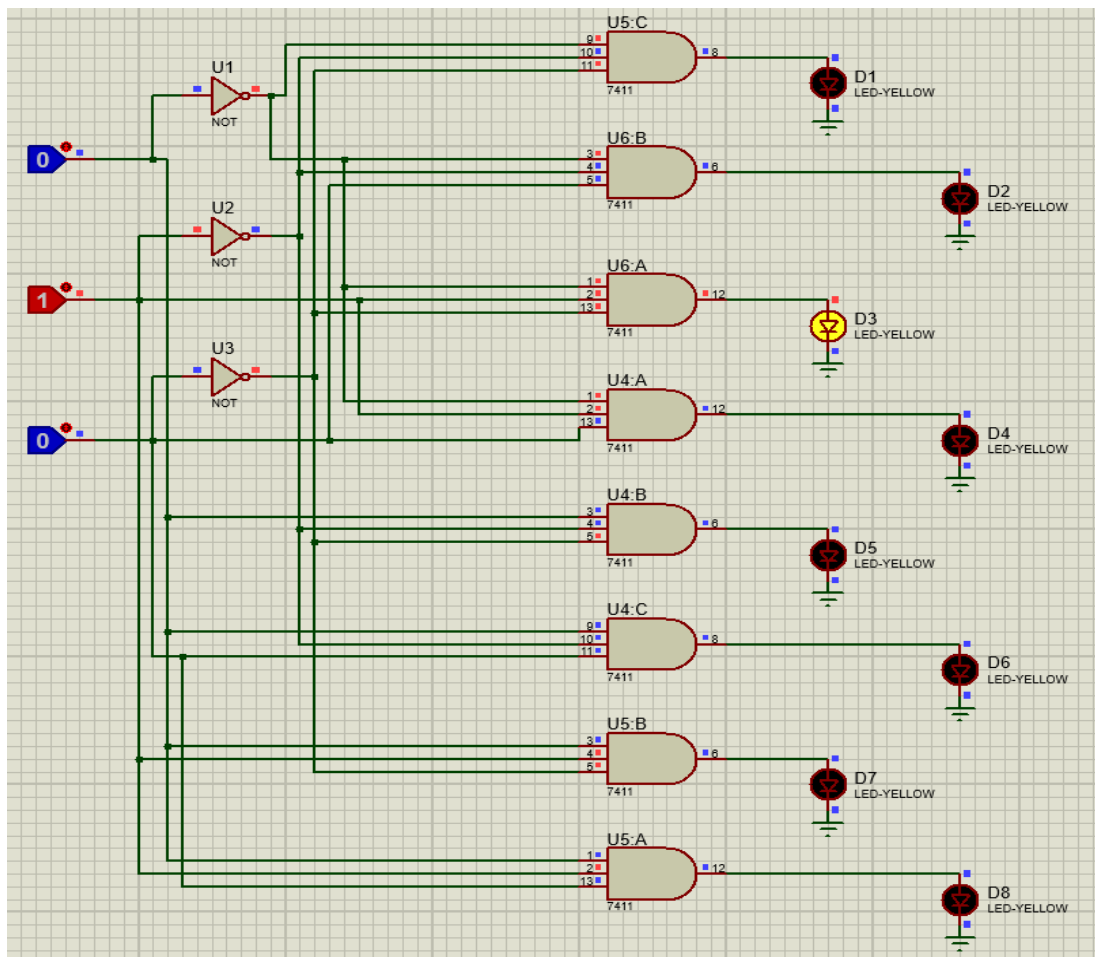
A	B	C	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Simulated Results:

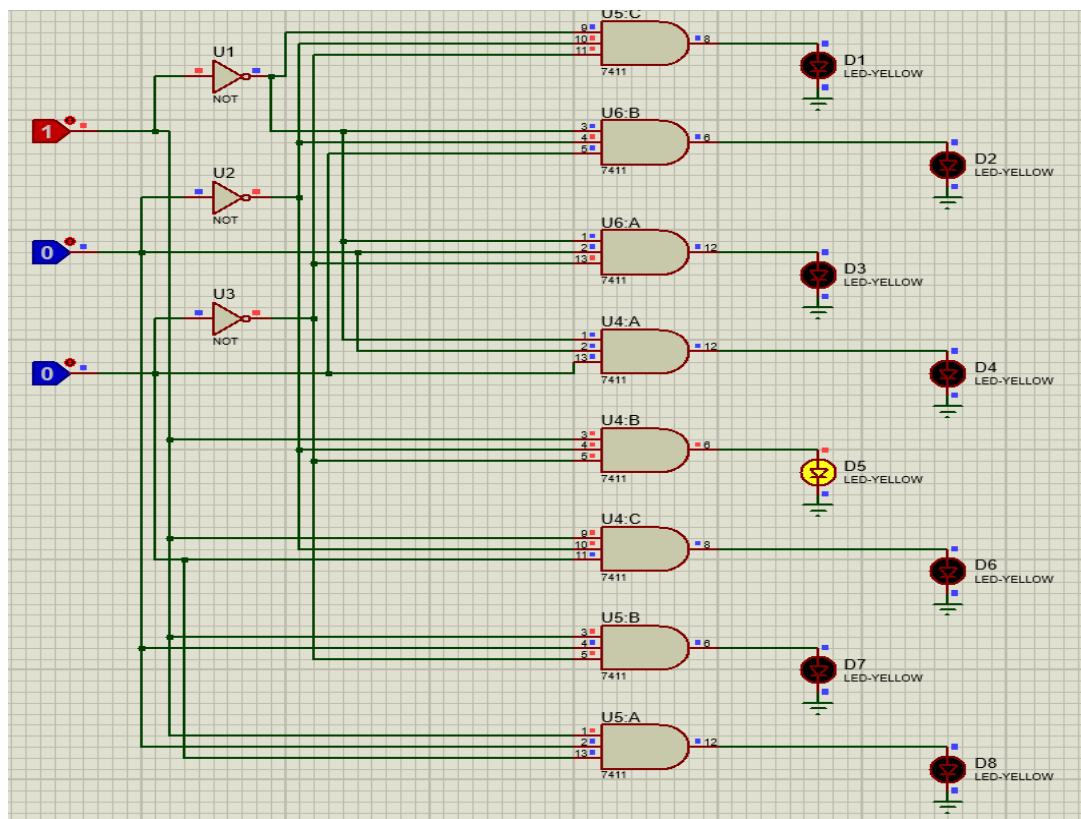
Input: A = 0, B = 0, C = 0 → Output: D₀ = 1



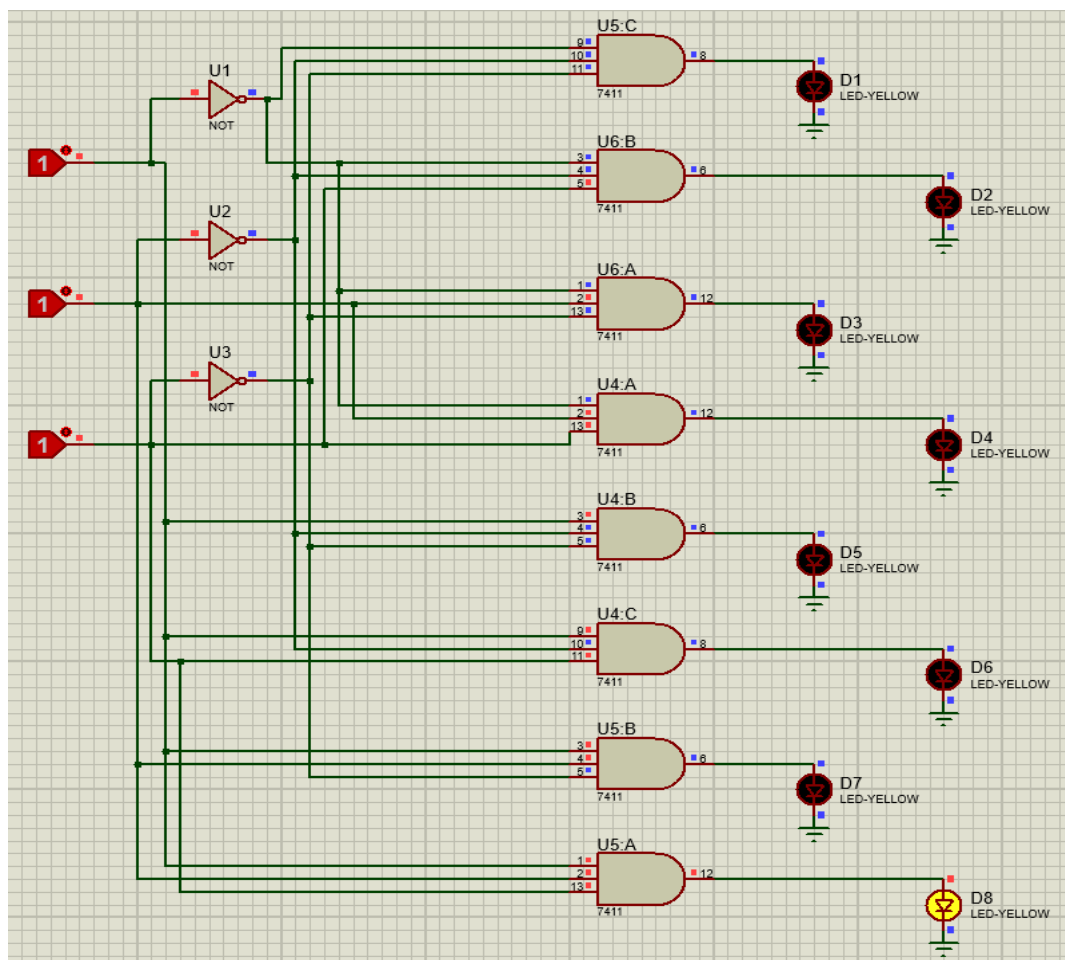
Input: A = 0, B = 1, C = 0 → Output: D₂ = 1



Input: A = 1, B = 0, C = 0 → Output: D₄ = 1



Input: A = 1, B = 1, C = 1 → Output: D₇ = 1



Conclusions:

This experiment successfully demonstrated the design and implementation of a decoder circuit using logic gates. The results from the truth table and Proteus simulation confirmed the circuit's functionality, as only one output line was activated for each unique input combination. Decoders are vital components of digital systems, serving purposes like memory addressing and data routing. Their ability to simplify complex tasks makes them crucial in a wide range of digital applications. Future experiments could explore more advanced decoder designs or investigate the integration of decoders with other combinational circuits to enhance performance in sophisticated digital environments.

THE END
